$S \longrightarrow D;S \mid D$

$D \longrightarrow id = E \mid E$

$E \longrightarrow T + E \mid T$

$T \longrightarrow F * T \mid F$

$F \longrightarrow (E) \mid id \mid num$

$S' \longrightarrow ;S \mid \varepsilon$

$S \longrightarrow DS'$

$E' \longrightarrow +E \mid \varepsilon$

$E \longrightarrow TE'$

$T' \longrightarrow *T \mid \varepsilon$

$T \longrightarrow FT'$

$D \longrightarrow id = E \mid E$

$F \longrightarrow (E) \mid id \mid num$

$P(S') = \{ ;, \varepsilon \}$

$P(S) = \{ id, (, num \}$

$P(E') = \{ +, \varepsilon \}$

$P(E) = \{ (, id, num \}$

$P(T') = \{ *, \varepsilon \}$

$P(T) = \{ (, id, num \}$

$P(D) = \{ id, (, num \}$

$P(F) = \{ (, id, num \}$

$\underset{A}{S'} \longrightarrow \underset{\alpha}{;} \underset{Z}{S} \mid \varepsilon$ $\quad S(S) = \{ \$,$

$S(S) \longrightarrow S(S')$

$S(S) = \{ \$ \}$

$S(S') = \{ \$ \}$

$$\underset{A}{E'} \longrightarrow \underset{\alpha}{+}\underset{Z}{E} \mid \varepsilon \qquad S(E) = S(E')$$

$$\underset{A}{D} \longrightarrow \underset{\alpha}{id =} \underset{Z}{E} \mid E$$

$$F \longrightarrow \underset{\alpha}{(} \underset{Z}{E} \underset{\beta}{)} \mid id \mid num$$

$$S(E) \longrightarrow S(D) \qquad S(E) \rightarrow \{),\ ;,\ \$\}$$

$$P())\longrightarrow \{)\} \qquad S(E') \rightarrow \{),\ ;,\ \$\}$$

$$D \longrightarrow \underset{\alpha}{} \underset{Z}{E}$$

$$S(E) \longrightarrow S(D)$$

$$S(D)$$

$$\underset{A}{S} \longrightarrow \underset{\alpha}{} \underset{Z}{D} \underset{\beta}{S'}$$

$$S(D) \rightarrow \{;,\ \$\}$$

$$S(T) \rightarrow \{+,\ ),\ ;,\ \$\}$$

$$\underset{A}{E} \longrightarrow \underset{\alpha}{} \underset{Z}{T} \underset{\beta}{E'}$$

$S(T') \longrightarrow \{+,), ;, \$\}$

$\underset{A}{T} \longrightarrow \underset{\alpha}{F}\ \underset{z}{T'}$

$S(T') \longrightarrow S(T)$

$S(F) \longrightarrow$

$T \longrightarrow \underset{\alpha}{}\ \underset{z}{F}\ \underset{B}{T'}$

$S(F) = \{*, +,), ;, \$\}$

	id	num	(	;	+	*	$	)	=
S	S → DS'	S → DS'	S → DS'						
S'				S' → ;S			S' → ε		
E'				E' → ε	E' → +E		E' → ε	E → ε	
E	E → TE'	E → TE'	E → TE'						
T'				T' → ε	T' → ε	T' → *T	T' → ε	T' → ε	
T	T → FT'	T → FT'	T → FT'						
F	F → id	F → num	F → (E)						
D	D → id=E	D → E	D → E						

S' ⟶ ; S | ε

S ⟶ DS'

E' ⟶ + E | ε

E ⟶ T E'

T' ⟶ * T | ε

T ⟶ F T'

D ⟶ id = E | E

F ⟶ (E) | id | num

Pila	cadena
$ S	id=num;num=id$
$ S'D	~~id~~=num;num=id$
$ S'E	id=num;num=id$
$ S'E'T	num;num=id$
$ S'E'T'F	num;num=id$
$ S'	~~;~~num=id$
$ S'S	num=id$
$ S'S'D	num=id$
$ S'S'E	num=id$
$ S'S'E'T	~~num~~=id$
$ S'S'E'T'F	=id$
$ S'S'E'T'	=id$

id = num ; num = id

1 2 3 4